

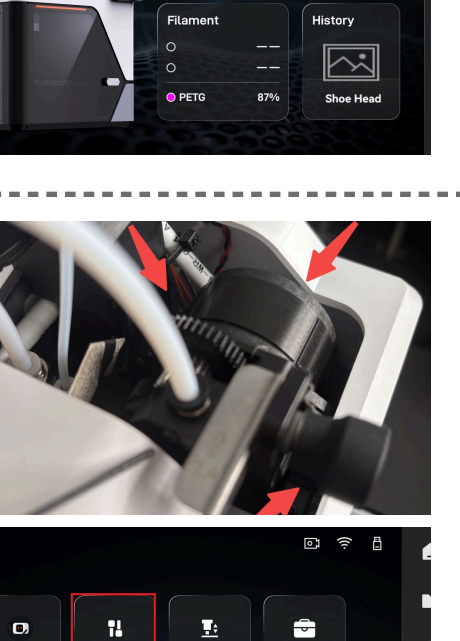
Plastic Hotend Unclogging

This SOP outlines how to clean plastic hotend clogs in the FibreSeeker 3, aiming to safely and effectively resolve clogging.

1. Tools:

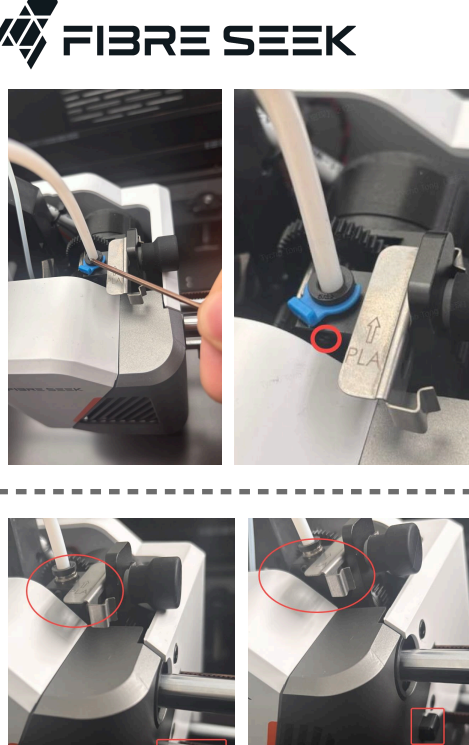
H3, H2.5, and H1.5 hex keys, an unclogging pin tool, and heat-resistant gloves.

2. Precautions:



Wear heat-resistant gloves to avoid burn accidents when you clean the nozzle. Before troubleshooting, check whether the filament is tangled.

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Step 1

Heat the Nozzle

Heat the right nozzle to the temperature specified for the corresponding material, for example, PETG at 250°C, then try extruding again. Lower temperatures may block extrusion.

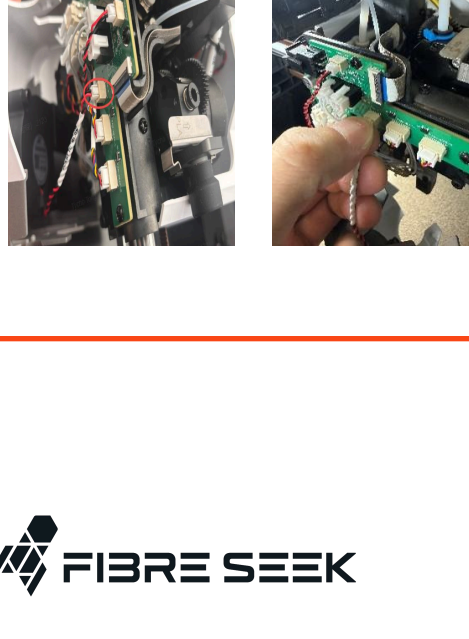


Step 2

Check Rocker Arm Screws, Extrude, and Inspect Gears

Tighten the rocker arm fastening screws and try extruding again. Tap the settings icon on the right side of the toolbar, then select "Control" - "Extruder". Set the temperature to approximately 250°C and the length to 50mm, then tap the down arrow to extrude. Observe whether the extruder gear is turning. If it turns properly, proceed to the next steps. If it does not turn, check whether the motor is operating. If the motor is not running, contact customer support for assistance.

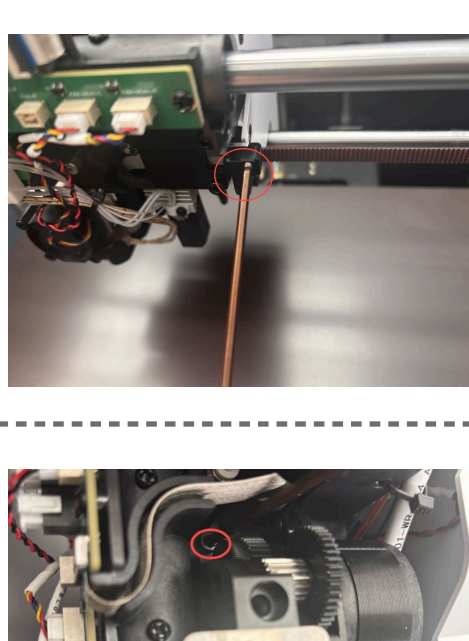
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Step 3

Remove the PTFE Tube

If material is still not extruding, manually remove the PTFE tube: Remove the blue retaining clip. (Press the black fitting connector all the way down and pull the tube out forcefully.) If the tube cannot be removed, use an H1.5 hex key to loosen the set screws and detach the fitting connector.



Step 4

Switch the Nozzle

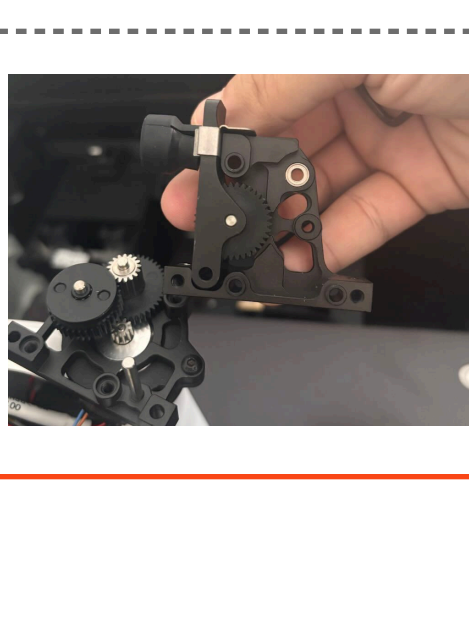
Push the nozzle to the left to switch it. This will cause the plastic hotend section to move upward. (You can observe changes in the set screw position in the circled area.)

Step 5

Check and Verify Filament

Manually pull out the filament left at the extruder (shown without the tube fitting). If the filament can be pulled out with minimal force, the extruder gear is worn. To verify again, use a smooth filament and reinsert it into the extruder. You should feel gear friction; the filament should be insertable with slight force. Ensure the rocker arm fastening screws are tightened. Refer to the Extruder Replacement Guide for further instructions.

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Step 6

Upload the Filament

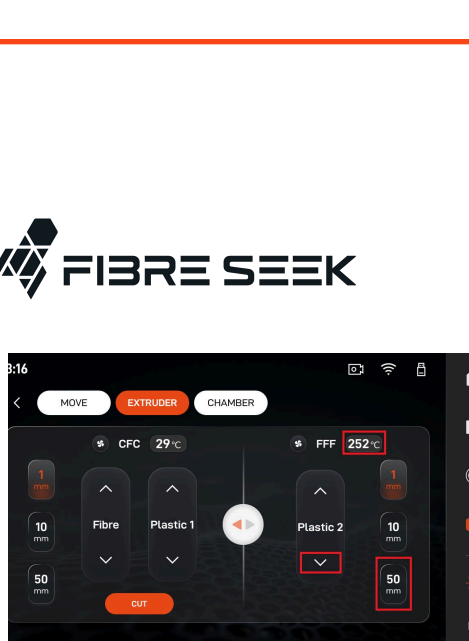
If the filament cannot be pulled out with slight force, click Unload Filament. If the material still cannot eject, dismantle the extruder for inspection. If the filament can be successfully unloaded, skip the extruder dismantling step and refer to Step 7.2 Nozzle Clogging.

Step 7.1

Remove the Extruder

Step 7.1.1

Power off the printer. Press the bottom clips on both sides of the front cover simultaneously, then pull the cover forward to remove it.



Step 7.1.2

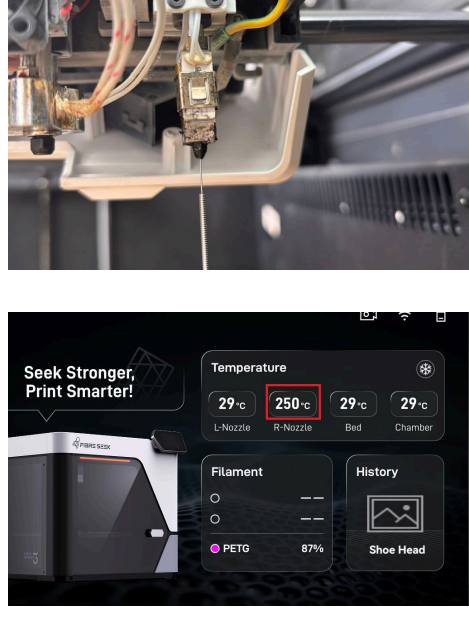
Pinch the latch on the fan at the "FAN-R" position on the mainboard and unplug it to remove the front cover of the toolhead.

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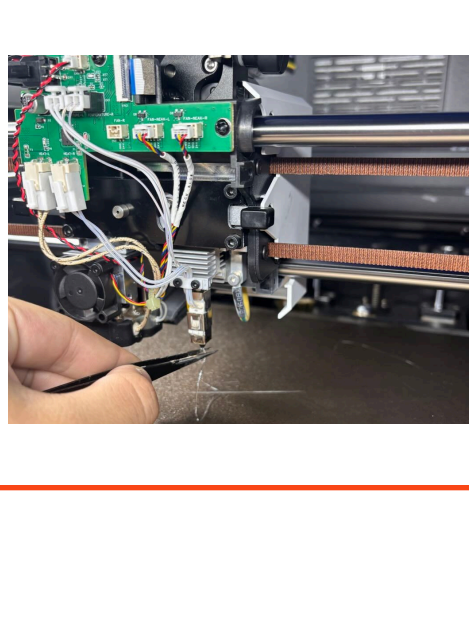
Step 7.1.3

Push leftward by hand to switch the nozzle. This will cause the plastic hotend section to move upward. (Skipping this step may prevent the extruder from being removed.)



Step 7.1.4

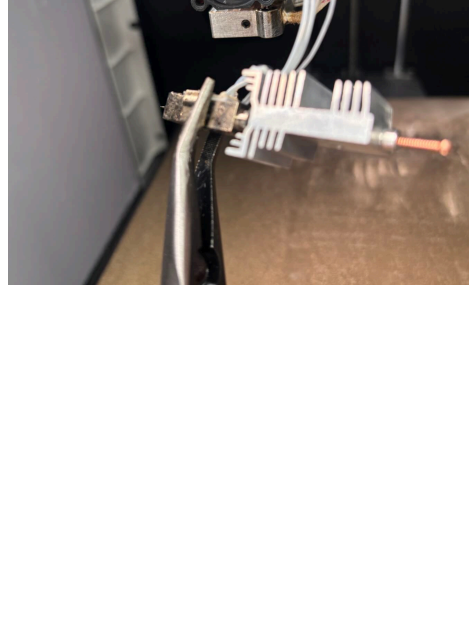
Use an H2.5 hex key to loosen the 2 fixing screws and take them out. Slightly pry the rear cover, wiggle the extruder left and right, and remove it.



Step 7.1.5

Use the H2.5 hex key to loosen the 3 fixing screws and disassemble the extruder.

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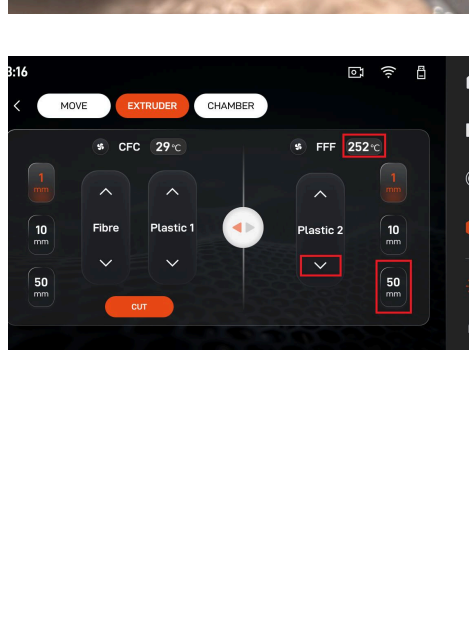
Step 7.1.6

Use a tool to gently pry it apart for easier disassembly.



Step 7.1.7

Check if the extruder is jammed or broken due to internal filament.



Step 7.1.8

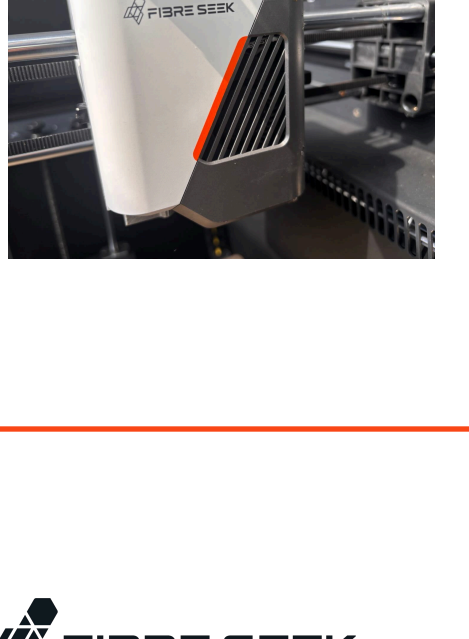
If filament is stuck, clean it thoroughly, then reassemble in order. Align the parts and press firmly until you hear a click to ensure no gaps.

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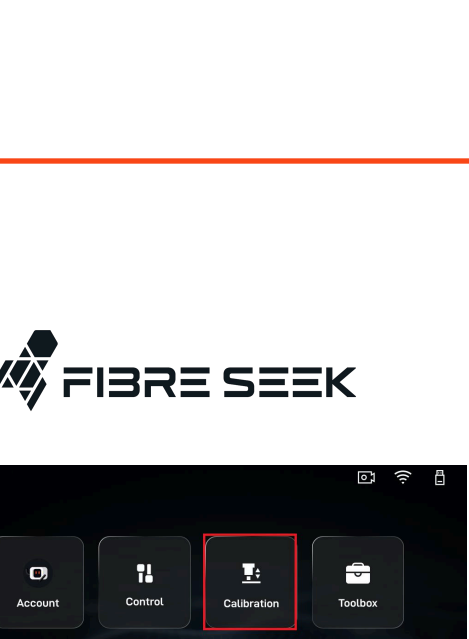
Step 7.1.9

Reinstall and tighten the fixing screws.



Step 7.1.10

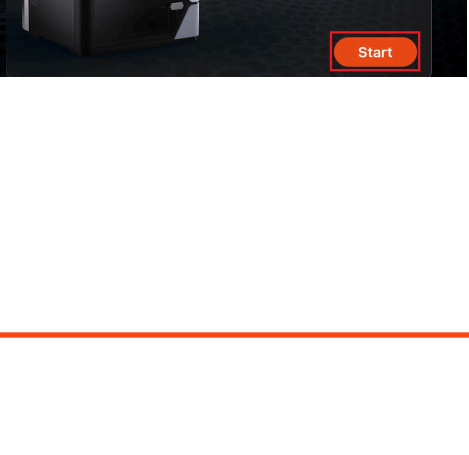
Insert filament and reextrude.



Step 7.1.11

Once normal extrusion is confirmed, reinstall the extruder.

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Step 7.1.12

After reinstalling the extruder, test extrusion to verify nozzle output. If no material exits, go to Step 7.2. Otherwise, run a normal print test.

Step 7.2

Nozzle Clogging

Step 7.2.1

Power off the printer. Press the bottom clips on both sides of the front cover simultaneously, then pull the cover forward to remove it.

Step 7.2.2

Pinch the latch on the fan at the "FAN-R" position on the mainboard and unplug it to remove the front cover of the toolhead.

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Step 7.2.3

Heat the nozzle, then use an unclogging pin tool to push through from the bottom (nozzle tip) upward. Repeat several times. If the needle moves smoothly, try extruding material.

Step 7.2.4

If extrusion succeeds, extrude several times to flush out residual debris and clean with tweezers. Once cleaned, run a print test. If no material comes out, proceed to the step below.

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Step 7.2.5

Use an H2.5 hex key to loosen 2 fixing screws and remove the plastic hotend assembly.

Step 7.2.6

Clean the internal channel with the unclogging pin tool. If that fails, replace the plastic hotend assembly (see the Replacement Guide).

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Step 7.2.7

If cleared (needle moves back and forth smoothly), reinstall the plastic hotend and try extruding again.

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Step 7.2.8

Plug the fan cable of the front cover back into the corresponding port on the mainboard. Make sure the two bottom clips align with their holes and snap into place.

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Step 7.2.9

After installation, manually calibrate the nozzle offset.

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